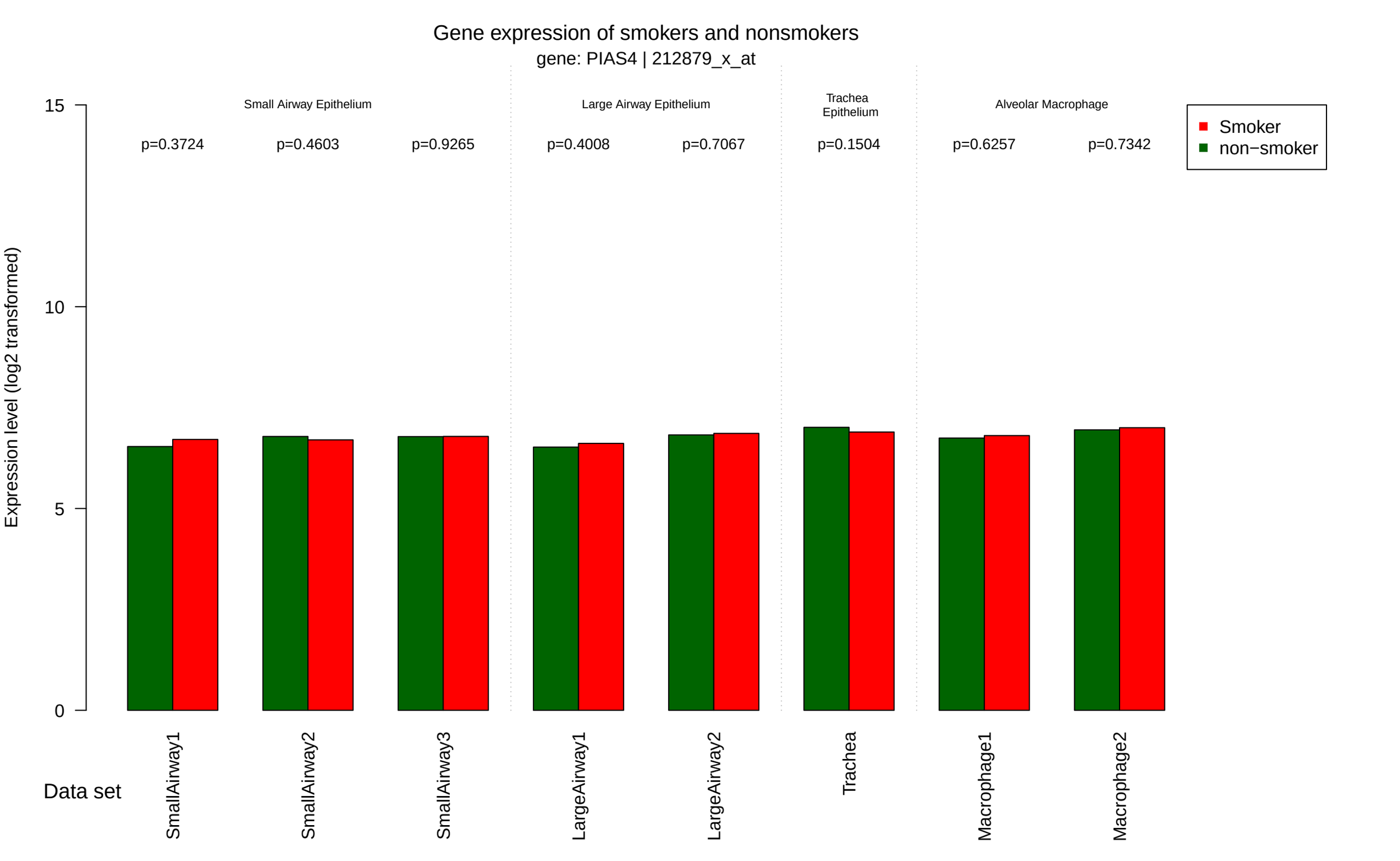
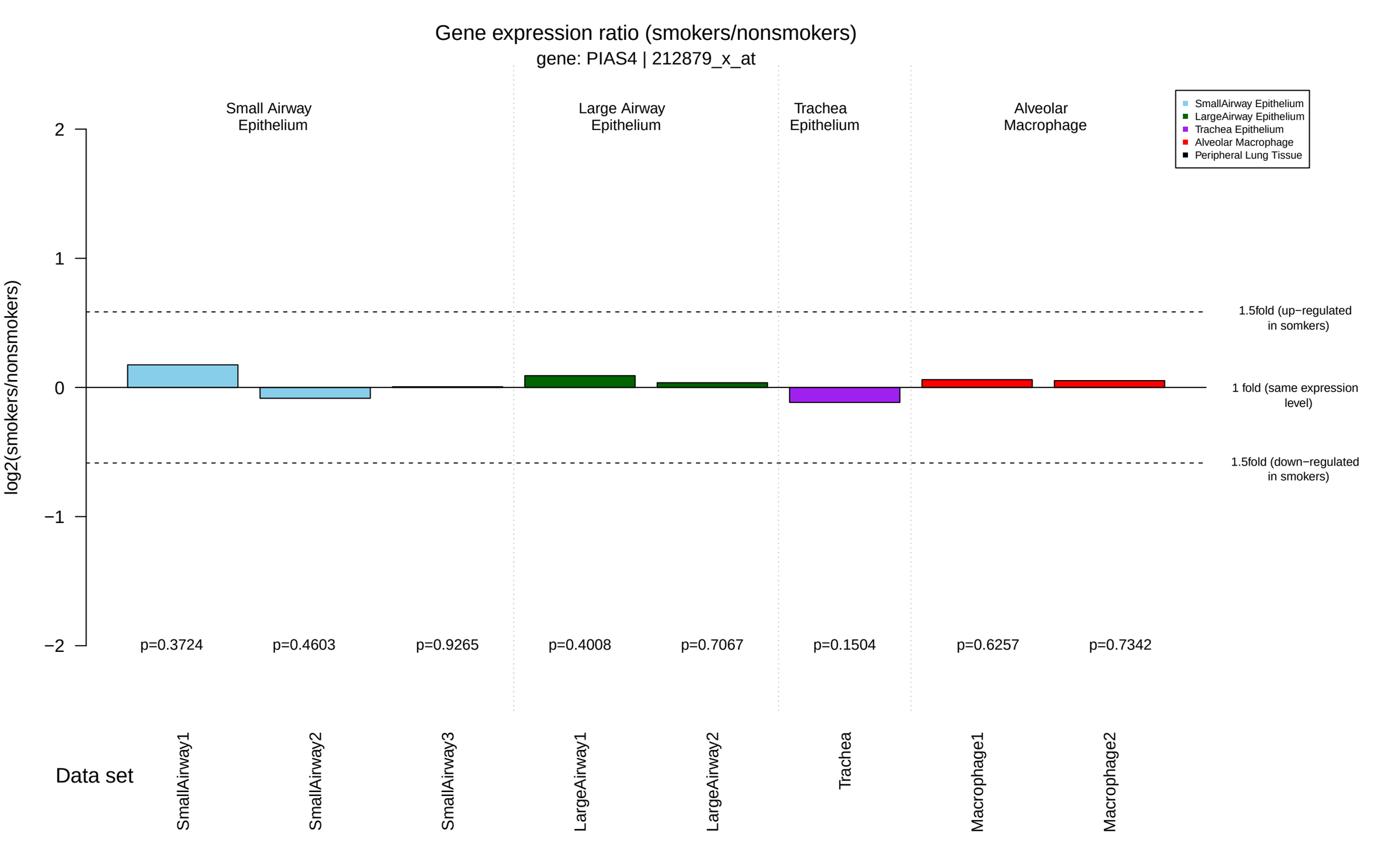
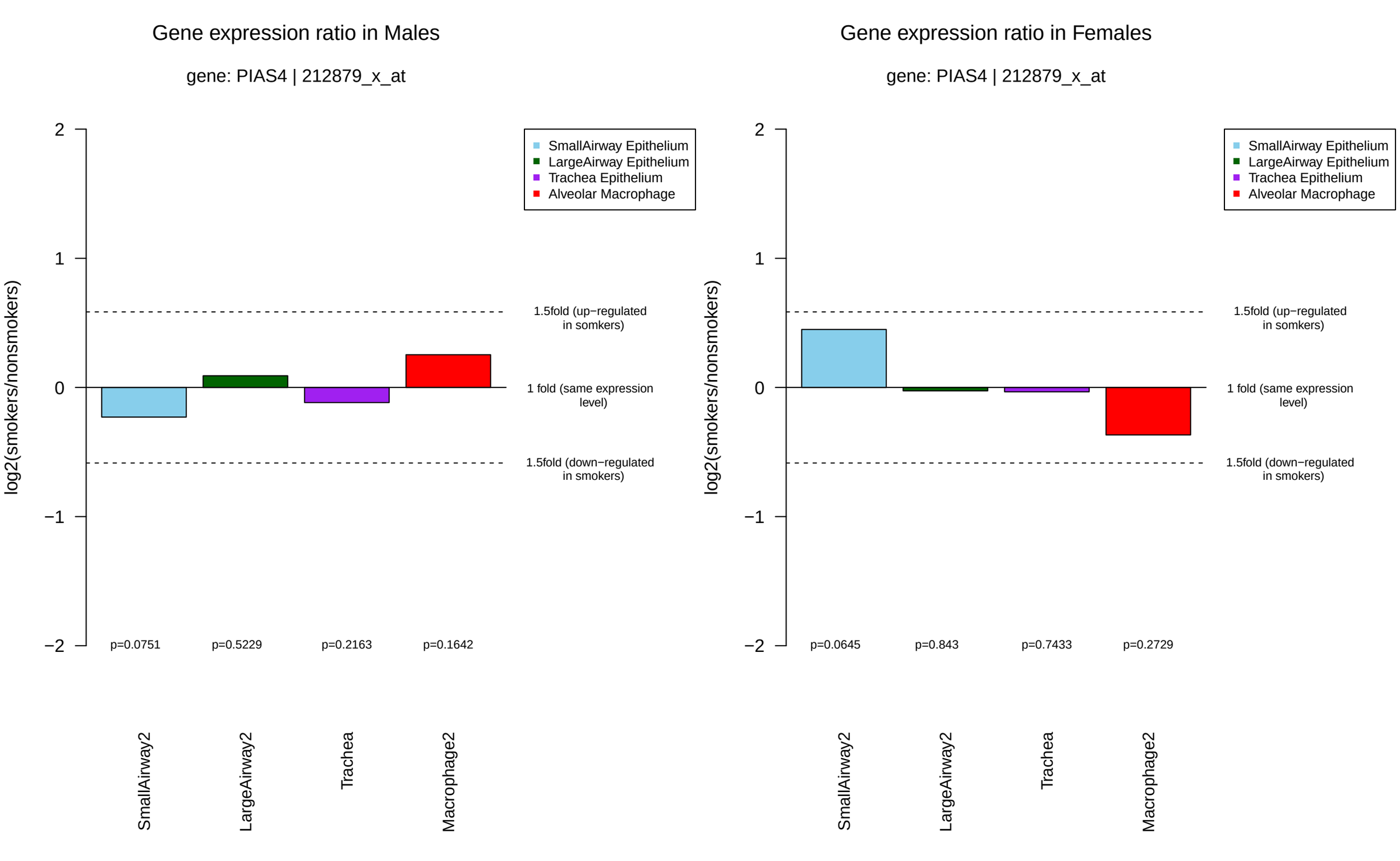
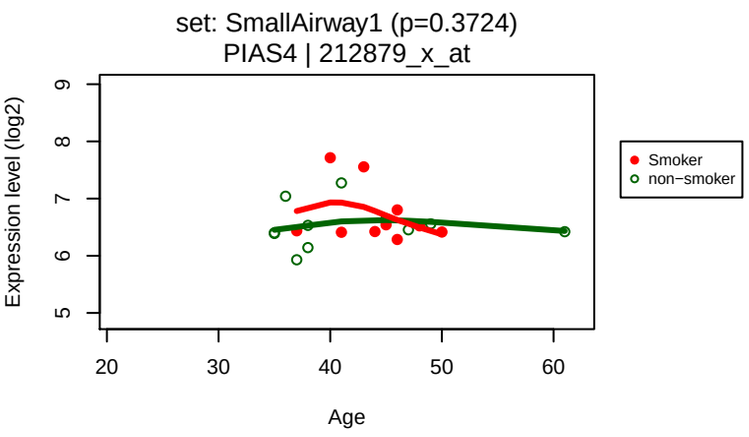
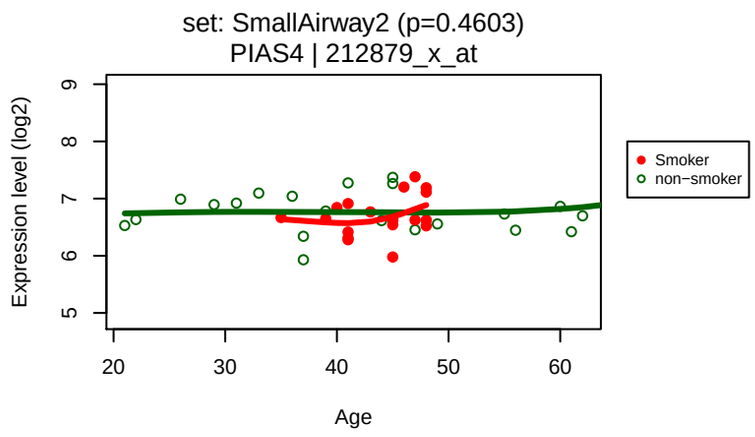
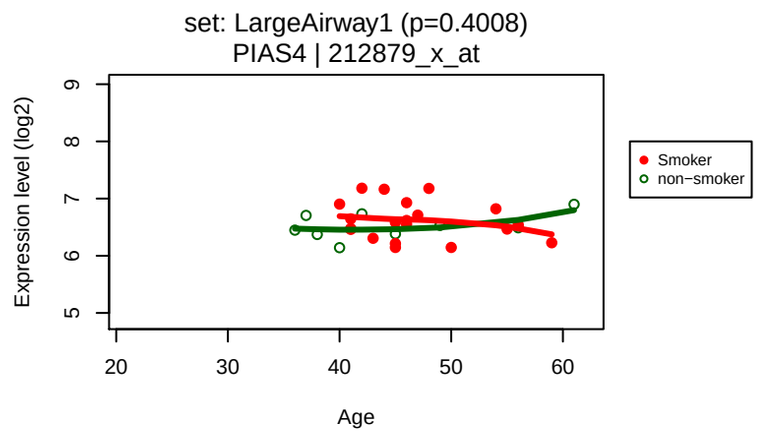

Gene expression of smokers and nonsmokers
gene: PIAS4 | 212879_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3724
p=0.4603
p=0.9265
p=0.4008
p=0.7067
p=0.1504
p=0.6257
p=0.7342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

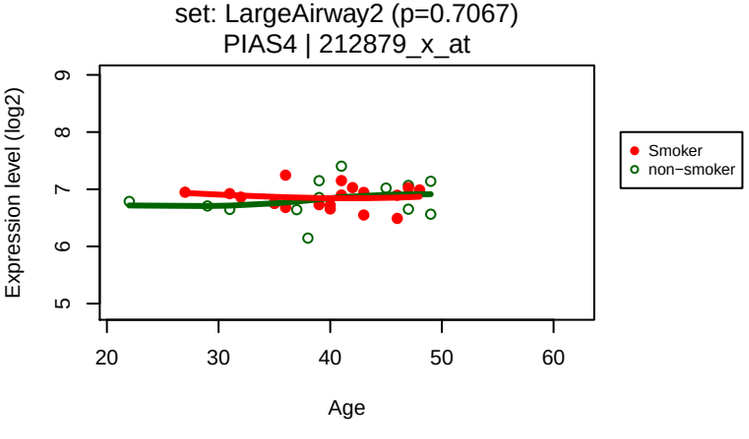

Gene expression ratio (smokers/nonsmokers)
gene: PIAS4 | 212879_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3724
p=0.4603
p=0.9265
p=0.4008
p=0.7067
p=0.1504
p=0.6257
p=0.7342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

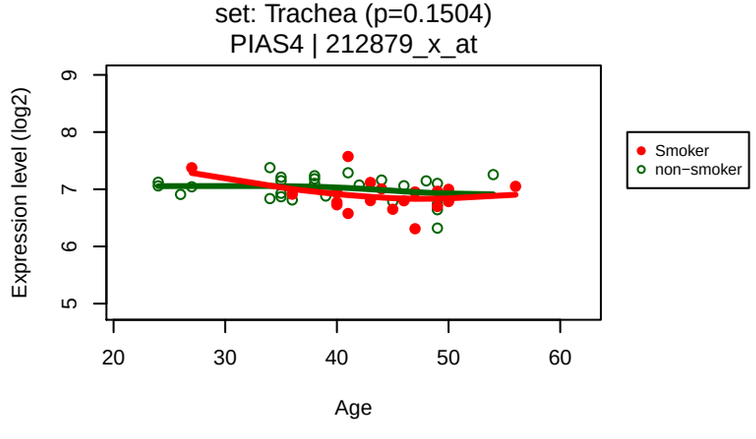

Gene expression ratio in Males
gene: PIAS4 | 212879_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0751
p=0.5229
p=0.2163
p=0.1642
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIAS4 | 212879_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0645
p=0.843
p=0.7433
p=0.2729
SmallAirway2
LargeAirway2
Trachea
Macrophage2

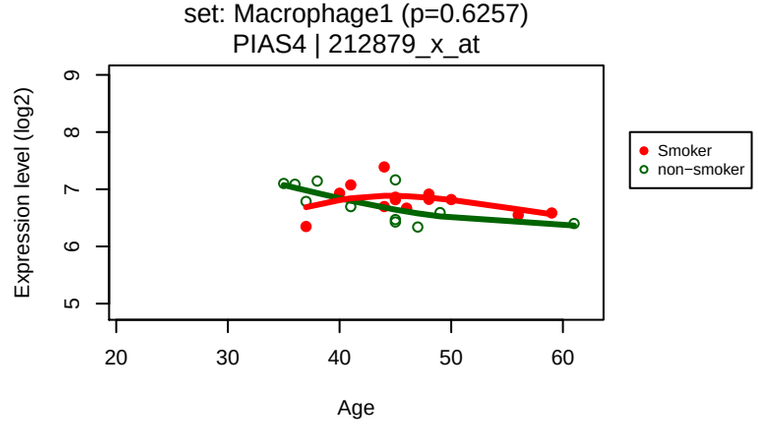

set: SmallAirway1 (p=0.3724)
PIAS4 | 212879_x_at
Expression level (log2)
Age
Smoker
non−smoker

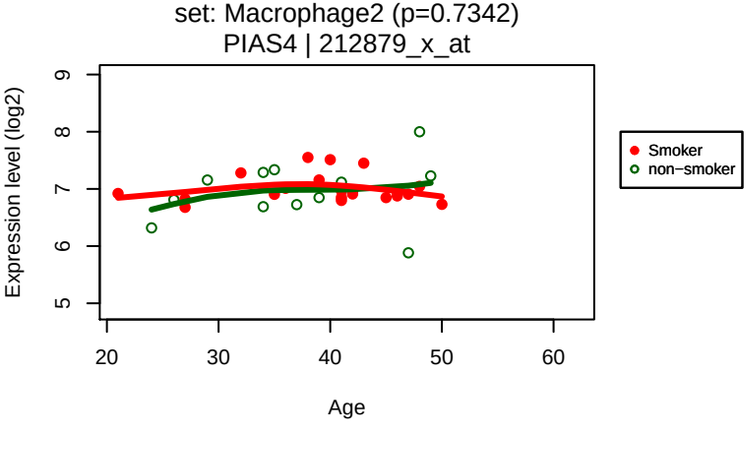

set: SmallAirway2 (p=0.4603)
PIAS4 | 212879_x_at
Expression level (log2)
Age
Smoker
non−smoker

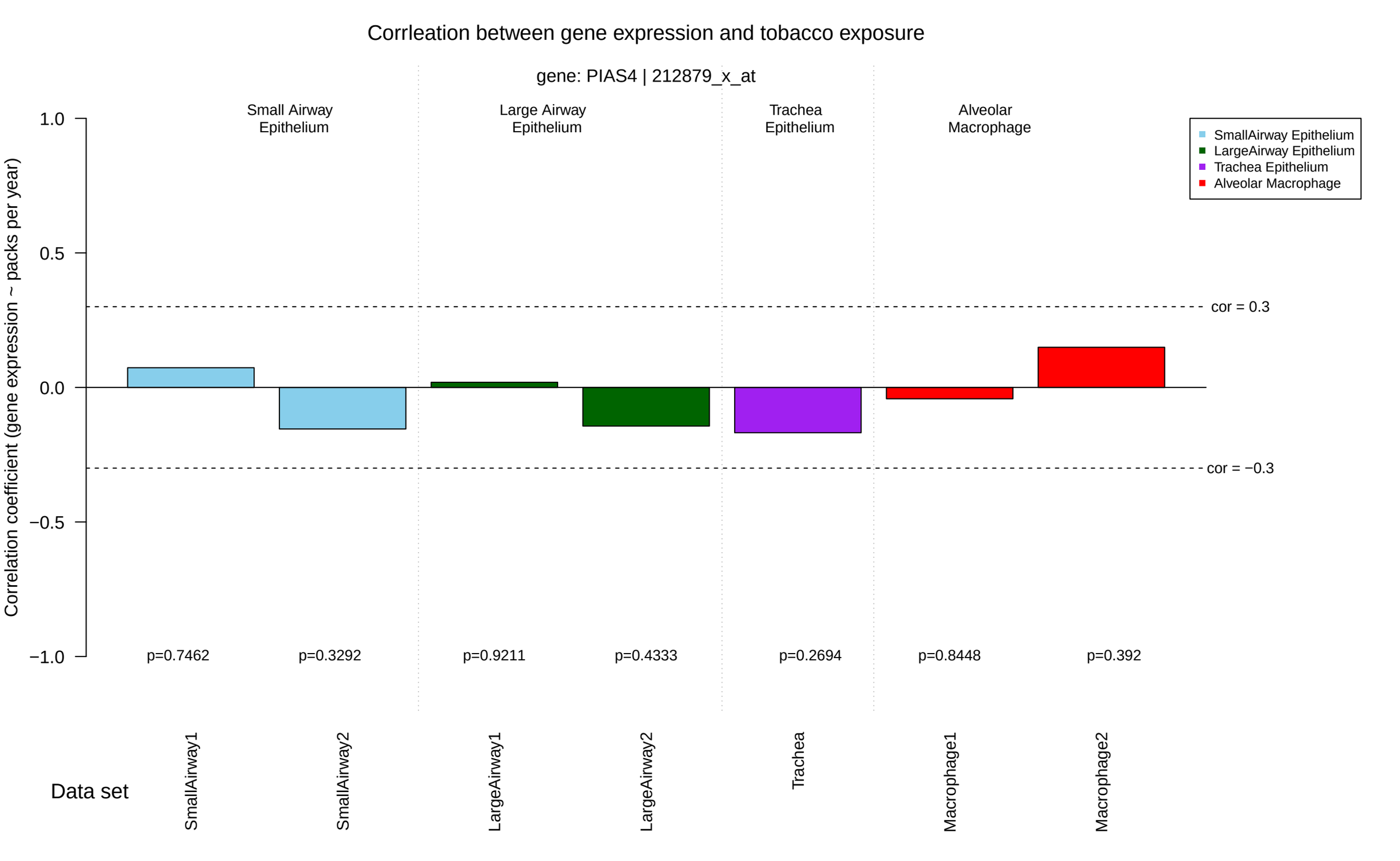

set: LargeAirway1 (p=0.4008)
PIAS4 | 212879_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7067)
PIAS4 | 212879_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1504)
PIAS4 | 212879_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6257)
PIAS4 | 212879_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7342)
PIAS4 | 212879_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIAS4 | 212879_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7462
p=0.3292
p=0.9211
p=0.4333
p=0.2694
p=0.8448
p=0.392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2